

1. Select any open government dataset and perform data exploration using importing, table functions, filtering, sorting, grouping, correlation analysis, and outlier detection. Present the results using suitable charts and publish the final analysis on an open-source platform.

SOLUTION:

Dataset Selected: World Development Indicators

For this analysis, we use the World Development Indicators (WDI) dataset. It contains information about countries such as GDP, population, life expectancy, and unemployment.

The analysis performs:

- Data importing
- Table functions
- Filtering
- Sorting
- Grouping
- Correlation analysis
- Outlier detection

Complete Python Code

```
import pandas as pd
df = pd.read_csv("world_development.csv")
print("First 5 rows:")
print(df.head())
print("\nDataset Shape:")
print(df.shape)
print("\nColumn Names:")
print(df.columns)
```

```

print("\nDataset Information:")

print(df.info())

print("\nStatistical Summary:")

print(df.describe())

print("\nMissing Values:")

print(df.isnull().sum())

print("\nCountries with GDP greater than 100000:")

high_gdp = df[df["GDP"] > 100000]

print(high_gdp)

print("\nTop 10 Countries by GDP:")

top_gdp = df.sort_values("GDP", ascending=False)

print(top_gdp[["Country", "GDP"]].head(10))

print("\nAverage GDP by Region:")

region_gdp = df.groupby("Region")["GDP"].mean()

print(region_gdp)

print("\nCorrelation Analysis:")

correlation = df[

    ["GDP", "Population", "Life_Expectancy", "Unemployment"]

].corr()

print(correlation)

Q1 = df["GDP"].quantile(0.25)

Q3 = df["GDP"].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR

upper_limit = Q3 + 1.5 * IQR

outliers = df[

    (df["GDP"] < lower_limit) |

    (df["GDP"] > upper_limit)

]

print("\nGDP Outliers:")

print(outliers[["Country", "GDP"]])

print("\nNumber of GDP Outliers:")

```

```
print(len(outliers))
```

Sample Output

Note: The exact output depends on the version of the WDI dataset you use.

First 5 rows:

	Country	Region	GDP	Population	Life_Expectancy	Unemployment
0	India	South Asia	3.57e+12	1428627663	67.7	7.2
1	China	East Asia	1.77e+13	1410710000	78.2	5.3
2	Japan	East Asia	4.21e+12	124516650	84.0	2.6
3	Germany	Europe	4.46e+12	83294633	81.0	3.0
4	United States	North America	2.77e+13	339996563	76.4	3.6

Dataset Shape

Dataset Shape:

(265, 6)

Column Names

```
Index([
    'Country',
    'Region',
    'GDP',
    'Population',
    'Life_Expectancy',
    'Unemployment'
], dtype='object')
```

Statistical Summary

	GDP	Population	Life_Expectancy	Unemployment
count	250.0000	2.500000e+02	250.000000	250.000000
mean	1.20e+12	3.20e+07	71.500000	6.200000
std	3.50e+12	1.40e+08	8.500000	5.100000
min	5.00e+07	1.000000e+05	50.000000	0.500000
max	2.77e+13	1.43e+09	84.500000	30.000000

Missing Values

```
Country      0
Region       0
GDP          15
Population   10
Life_Expectancy  8
Unemployment  12
dtype: int64
```

Filtering Output

Countries with GDP greater than 100000:

```
Country      GDP
China      1.77e+13
United States  2.77e+13
Japan      4.21e+12
Germany     4.46e+12
India      3.57e+12
...
```

Sorting Output

Top 10 Countries by GDP:

```
Country      GDP
1  United States  2.77e+13
2    China  1.77e+13
3    Japan  4.21e+12
4  Germany  4.46e+12
5    India  3.57e+12
...
```

Grouping Output

Average GDP by Region:

```
Region
East Asia      2.45e+12
```

Europe	1.85e+12
South Asia	9.20e+11
North America	8.40e+12
Africa	1.10e+11

dtype: float64

Correlation Output

Correlation Analysis:

	GDP	Population	Life_Expectancy	Unemployment
GDP	1.00	0.72	0.35	-0.12
Population	0.72	1.00	0.18	0.05
Life_Expectancy	0.35	0.18	1.00	-0.30
Unemployment	-0.12	0.05	-0.30	1.00

Interpretation: GDP and population have a relatively strong positive correlation of about 0.72. This indicates that countries with larger populations tend to have higher GDP in this dataset. Correlation does not by itself establish causation.

Outlier Detection Output

GDP Outliers:

	Country	GDP
1	United States	2.77e+13
2	China	1.77e+13
3	Japan	4.21e+12

Number of GDP Outliers: 3

5 Key Findings

1. GDP differs significantly between countries, with a small number of countries having exceptionally high GDP.
2. Population and GDP show a positive correlation, indicating that highly populated countries often have larger economies.
3. Life expectancy has a moderate relationship with GDP, suggesting differences in economic development may be associated with differences in life expectancy.
4. GDP contains outliers, mainly representing countries with exceptionally large economies.

6. Regional grouping shows significant differences in average GDP, highlighting economic differences between regions.

Recommendations

1. Countries with lower GDP should focus on economic development and infrastructure investment.
2. Governments should improve healthcare and education to support better life expectancy and human development.
3. High-GDP countries should continue investing in sustainable economic growth.
4. Policymakers should consider regional economic differences when designing development programs.
5. Outlier countries should be analyzed separately when comparing economic indicators to avoid misleading conclusions.

Conclusion

The World Development Indicators dataset was successfully explored using Python and Pandas. Data importing, table functions, filtering, sorting, grouping, correlation analysis, and outlier detection were performed. The analysis identified important relationships between GDP, population, life expectancy, and unemployment.

The final Python notebook can be published on GitHub as an open-source project so that the analysis and code can be accessed and reproduced by others.

2. Select any open environmental dataset such as air quality, rainfall, temperature, or pollution data. Analyze time-related patterns, identify correlations and outliers, create geographical visualizations/maps, and present the findings using an interactive visualization tool.

SOLUTION:

Dataset Selected: Air Quality Dataset

For this analysis, we use an open air-quality dataset containing measurements such as:

- Date
- City
- PM2.5
- PM10
- NO
- SO
- CO
- O

- Temperature
- Humidity
- Latitude
- Longitude

The analysis includes time-based patterns, correlation analysis, outlier detection, geographical mapping, and interactive visualization.

Complete Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import folium

df = pd.read_csv("air_quality.csv")
print("First 5 rows:")
print(df.head())
print("\nDataset Shape:")
print(df.shape)
print("\nColumn Names:")
print(df.columns)
print("\nMissing Values:")
print(df.isnull().sum())
df["Date"] = pd.to_datetime(df["Date"])
df = df.sort_values("Date")
print("\nData after sorting by date:")
print(df[["Date", "City", "PM2.5", "PM10"]].head())
df["Year"] = df["Date"].dt.year
df["Month"] = df["Date"].dt.month
monthly_pm25 = df.groupby("Month")["PM2.5"].mean()
print("\nAverage PM2.5 by Month:")
print(monthly_pm25)
yearly_pm25 = df.groupby("Year")["PM2.5"].mean()
print("\nAverage PM2.5 by Year:")
```

```

print(yearly_pm25)

city_pollution = df.groupby("City")[["PM2.5", "PM10"]].mean()

print("\nAverage Pollution by City:")

print(city_pollution)

correlation = df[
    ["PM2.5", "PM10", "NO2", "SO2", "CO", "O3",
     "Temperature", "Humidity"]
].corr()

print("\nCorrelation Matrix:")

print(correlation)

Q1 = df["PM2.5"].quantile(0.25)
Q3 = df["PM2.5"].quantile(0.75)
IQR = Q3 - Q1
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR
outliers = df[
    (df["PM2.5"] < lower_limit) |
    (df["PM2.5"] > upper_limit)
]

print("\nPM2.5 Outliers:")

print(outliers[["Date", "City", "PM2.5"]])

print("\nNumber of PM2.5 Outliers:")

print(len(outliers))

city_location = df.groupby("City").agg({
    "Latitude": "mean",
    "Longitude": "mean",
    "PM2.5": "mean"
}).reset_index()

m = folium.Map(
    location=[
        city_location["Latitude"].mean(),
        city_location["Longitude"].mean()
    ],

```



```

        zoom_start=5
    )
    for _, row in city_location.iterrows():
        folium.CircleMarker(
            location=[row["Latitude"], row["Longitude"]],
            radius=8,
            popup=(
                f"City: {row['City']}<br>"
                f"Average PM2.5: {row['PM2.5']:.2f}"
            ),
            fill=True
        ).add_to(m)
    m.save("air_quality_map.html")
    print("\nInteractive geographical map saved as air_quality_map.html")

```

Sample Output

The exact values depend on the open dataset used.

Dataset Information

First 5 rows:

	Date	City	PM2.5	PM10	NO2	SO2	CO	O3
0	2023-01-01	Chennai	45.2	72.4	21.3	10.2	0.8	32.5
1	2023-01-02	Chennai	51.4	80.1	24.2	11.5	0.9	30.2
2	2023-01-03	Mumbai	65.3	95.2	31.4	14.2	1.1	27.4
3	2023-01-04	Delhi	89.5	130.5	45.2	19.3	1.6	21.3
4	2023-01-05	Kolkata	72.1	105.3	35.2	16.4	1.2	25.7

Dataset Shape

Dataset Shape:

(1000, 9)

Missing Values

Date 0

City 0

PM2.5	10
PM10	8
NO2	12
SO2	5
CO	7
O3	6
Temperature	4
Humidity	3

dtype: int64

Time-Based Analysis

Average PM2.5 by Month

Month

1	82.45
2	76.32
3	68.21
4	54.73
5	48.62
6	42.31
7	38.54
8	40.21
9	45.62
10	58.43
11	71.24
12	84.53

Observation: PM2.5 levels are higher during some months and lower during others, showing a clear seasonal pattern.

Average PM2.5 by City

City

Chennai	45.32
Mumbai	58.64
Delhi	92.45

Kolkata 76.21

Bangalore 39.84

Observation: Delhi has the highest average PM2.5 among the example cities, while Bangalore has the lowest.

Correlation Analysis

	PM2.5	PM10	NO2	SO2	CO	O3
PM2.5	1.00	0.82	0.74	0.61	0.69	-0.32
PM10	0.82	1.00	0.68	0.57	0.64	-0.28
NO2	0.74	0.68	1.00	0.55	0.72	-0.21
SO2	0.61	0.57	0.55	1.00	0.51	-0.18
CO	0.69	0.64	0.72	0.51	1.00	-0.24
O3	-0.32	-0.28	-0.21	-0.18	-0.24	1.00

Observation: PM2.5 and PM10 have a strong positive correlation, meaning they tend to increase together.

Outlier Detection

PM2.5 Outliers:

Date	City	PM2.5
2023-12-15	Delhi	185.4
2023-12-21	Delhi	201.7
2024-01-05	Kolkata	172.3
2024-01-12	Delhi	215.8

Number of PM2.5 Outliers: 4

These unusually high values may indicate severe pollution episodes or unusual environmental conditions and should be investigated.

Key Findings

- 1. PM2.5 shows seasonal variation, with pollution levels changing significantly across different months.
- 2. Some cities have substantially higher pollution levels than others, indicating geographical differences in air quality.

3. PM2.5 and PM10 have a strong positive correlation, showing that particulate pollutants tend to increase together.
4. Outliers represent extreme pollution events, which may require investigation by environmental authorities.
5. Geographical mapping makes pollution hotspots easier to identify, allowing authorities to focus monitoring and pollution-control measures on high-risk areas.

Recommendations

1. Increase air-quality monitoring during months with historically high pollution.
2. Implement stricter controls on vehicle and industrial emissions in high-pollution cities.
3. Investigate extreme pollution outliers to determine their causes.
4. Use geographical pollution maps to identify and monitor pollution hotspots.
5. Promote public awareness and provide health warnings when pollution levels become dangerously high.

Conclusion

The environmental dataset was analyzed using Python and Pandas to identify time-based pollution patterns, correlations, and outliers. An interactive Folium geographical map was also created to visualize pollution levels across different locations. The analysis can help environmental authorities identify pollution hotspots, seasonal changes, and extreme pollution events and take appropriate preventive measures.